

Patient Shortage in Histopathology: Within-Patient Directions Locate the Missing Patient Span but Recover Only a Sixth of the Whitening Headroom

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September 21, 2026

Abstract

Whitening along the between-patient eigenbasis of a five-patient cohort recovers 2.26 points on TCGA-UT, while whitening along the training pool’s basis recovers 4.92. The preceding study attributed almost all of this difference to directions the cohort cannot see: its rank-120 basis holds 70 percent of the pool’s between-patient variance, and correcting the cohort’s eigenvalues is worth only 0.08 points. This report asks whether the patch residuals of the same patients reveal the missing directions that their means cannot. The cohort’s between-patient covariance is extended by its within-patient patch covariance projected off the cohort basis, with the added trace set to a fraction $s \in \{0.1, 0.3, 1, 3\}$ of the between-patient trace and s tuned on validation; a control keeps the same added eigenvalues but places them on random directions in the same complement. A pre-fit check confirms that the within-patient complement locates the missing span: its leading 120 directions hold 14.2 percent of the pool’s between-patient variance, against 20.3 percent for the best 120-dimensional subspace of the complement and 1.5 percent for random directions. The accuracy gain is much smaller. On TCGA-UT the tuned arm gains 0.45 points over cohort whitening at five patients (95% interval: 0.27 to 0.63), which is 17 percent of the 2.66-point gap to pool-basis whitening and fails the prespecified one-point criterion; the gain is direction-specific, since the random control gains 0.02 points, and it vanishes at ten and twenty patients. On BRACS the tuned arm gains 0.45 points (−0.35 to 1.21), not distinguishable from zero. Reaching the missing span is therefore not sufficient. The within-patient directions cover most of it, but they weight it by patch-level rather than patient-level variance, and the cohort offers no estimate of the latter along those directions.

1 Motivation

Three cohort-only remedies for patient shortage stop near the same ceiling on TCGA-UT. Whitening the features along the eigenbasis of the cohort’s own between-patient covariance gains 2.26 percentage points at five patients per class (Queisler, 2026e), directional shrinkage of the class centres along the same basis gains 1.29 points (Queisler, 2026d), and training against the class-centre uncertainty built from that covariance gains 2.34 points (Queisler, 2026c). Whitening along the between-patient basis of the full training pool gains 4.92 points at the same patient count, but it reads all training patients and is a mechanism reference rather than a remedy (Queisler, 2026e).

Queisler (2026f) split the 2.66-point gap between cohort-basis and pool-basis whitening into two parts. Replacing the cohort’s eigenvalues by the pool’s between-patient variance along the same directions gains 0.08 points; the remaining 2.58 points belong to directions outside the cohort basis. That basis has rank at most $C(G - 1)$, which is 120 on TCGA-UT at five patients, and it holds 70 percent of the pool’s between-patient variance. The same study found that whitening along the top within-patient principal components of matched rank performs as well as cohort whitening, and that the two subspaces overlap by 0.45, ten times the random

expectation. Patch-level variation within patients and the directions along which patient means differ are therefore partly the same directions.

This observation suggests a cohort-only way to widen the basis. The cohort’s patient means give only $C(G-1)$ degrees of freedom, but its patch residuals around those means give $CG(m-1)$, where m is the number of patches per patient; at five patients with 32 patches this is 4,650 on TCGA-UT, more than the feature dimension. If the within-patient covariance has high variance along the between-patient directions that the cohort’s means miss, it can supply those directions without any extra patient. Adding a second covariance to an unstable one follows the idea of regularized discriminant analysis, which blends class covariances with a pooled covariance through one mixing weight (Friedman, 1989), and of linear shrinkage toward a structured target (Ledoit and Wolf, 2004). Here the target is not chosen for conditioning but because it may point along the missing patient directions.

The question is therefore: *do the patch residuals of the same G patients reveal between-patient directions that their G patient means cannot, so that adding them to the cohort basis recovers part of the pool-basis whitening headroom?* Cohort whitening hosts the test, as in Queisler (2026f). The answer is judged at five patients per class, separately on each dataset, against the one-point practical threshold of this series. A random-direction control with the same added eigenvalues separates a gain from the specific directions from a gain from contracting more of the feature space.

2 Estimator model

Write the patch features of class c as $\mathbf{x}_{ik} = \boldsymbol{\mu}_c + \mathbf{u}_i + \mathbf{e}_{ik}$, with class centre $\boldsymbol{\mu}_c$, patient offset $\mathbf{u}_i$, and within-patient residual $\mathbf{e}_{ik}$ (Queisler, 2026b). A cohort of G patients per class estimates the between-patient covariance by pooling the deviations of its patient means $\mathbf{m}_{ci}$ from their class means over all C classes,

$$\hat{\boldsymbol{\Sigma}}_B = \frac{1}{C(G-1)} \sum_{c=1}^C \sum_{i=1}^G (\mathbf{m}_{ci} - \hat{\boldsymbol{\mu}}_c)(\mathbf{m}_{ci} - \hat{\boldsymbol{\mu}}_c)^\top = \mathbf{U}^\top \text{diag}(\boldsymbol{\lambda}) \mathbf{U}, \quad (1)$$

where the $r_B \leq C(G-1)$ rows of $\mathbf{U}$ are the eigenvectors with non-negligible eigenvalues $\boldsymbol{\lambda}$. Cohort whitening transforms every feature vector by

$$\mathbf{x} \mapsto \mathbf{x}(\mathbf{I} + \kappa \mathbf{B}^\top \text{diag}(\boldsymbol{\nu}) \mathbf{B})^{-1/2}, \quad (2)$$

with basis $\mathbf{B} = \mathbf{U}$ and spectrum $\boldsymbol{\nu} = \boldsymbol{\lambda}$; the transform shrinks the component along each basis row $\mathbf{b}_j$ by $(1 + \kappa \nu_j)^{-1/2}$ and leaves the orthogonal complement unchanged (Queisler, 2026e).

The within-patient covariance is formed from the patch deviations around each patient’s own mean, $\mathbf{x}_{ik} - \mathbf{m}_{ci}$, pooled over all patients and classes. Let $\mathbf{P}_\perp = \mathbf{I} - \mathbf{U}^\top \mathbf{U}$ project off the cohort basis. The eigen-decomposition of the projected within-patient covariance,

$$\mathbf{P}_\perp \hat{\boldsymbol{\Sigma}}_W \mathbf{P}_\perp = \mathbf{V}^\top \text{diag}(\boldsymbol{\omega}) \mathbf{V}, \quad (3)$$

gives the *within-patient complement*: orthonormal directions $\mathbf{V}$ that are orthogonal to $\mathbf{U}$ by construction, ordered by the patch-level variance $\boldsymbol{\omega}$ along them. Its rank is at most $\min\{d - r_B, CG(m-1)\}$, which fills the whole complement of the cohort basis on TCGA-UT and reaches about a thousand directions on BRACS. The expanded covariance adds this complement to the cohort’s between-patient covariance at a controlled total size,

$$\hat{\boldsymbol{\Sigma}}_s = \hat{\boldsymbol{\Sigma}}_B + s \tau \mathbf{P}_\perp \hat{\boldsymbol{\Sigma}}_W \mathbf{P}_\perp, \quad \tau = \frac{\text{tr} \hat{\boldsymbol{\Sigma}}_B}{\text{tr}(\mathbf{P}_\perp \hat{\boldsymbol{\Sigma}}_W \mathbf{P}_\perp)}. \quad (4)$$

The normalization τ makes the added trace equal to $s \operatorname{tr} \hat{\Sigma}_B$, so s is the size of the added part relative to the between-patient part, independent of the scale of the patch residuals. The added directions keep their own relative eigenvalues ω , and the cohort block keeps the cohort spectrum λ unchanged, because Queisler (2026f) found that flattening the cohort spectrum costs 0.92 points. The expanded arm whitens by Equation 2 with $\mathbf{B} = [\mathbf{U}; \mathbf{V}]$ and $\nu = [\lambda; s\tau\omega]$. With $s = 0$ it reduces to cohort whitening.

The *random-complement control* replaces $\mathbf{V}$ by the same number of random orthonormal directions orthogonal to $\mathbf{U}$ and keeps the eigenvalues $s\tau\omega$ in the same order. It contracts the same total amount of variance, spread with the same profile, in the same complement, but along directions that carry no information from the data. A gain of the within-patient arm over this control is therefore a gain from the specific directions, not from contracting more dimensions. On TCGA-UT the random directions span the same complement as $\mathbf{V}$ and differ only in which directions receive the large eigenvalues.

One further detail keeps the comparison with cohort whitening fair. The scale κ is tuned as a grid factor divided by the mean eigenvalue (Queisler, 2026e). Appending many small eigenvalues lowers the mean and would therefore whiten the cohort block harder at the same factor. The grid factors are rescaled by the ratio of the expanded mean eigenvalue to the cohort mean eigenvalue, so that every factor applies the same κ to the cohort block as it does in cohort whitening, and only the added block changes with s .

3 Design

The study keeps the setting of Queisler (2026b) on TCGA-UT and of Queisler (2026a) on BRACS. TCGA-UT contributes 30 cancer types and BRACS its seven lesion classes, both with frozen 2,560-dimensional Virchow2 features (Zimmermann et al., 2024), three fixed patient-disjoint splits, and multinomial logistic regression whose regularization strength is selected on validation patient-macro balanced accuracy. For each split and each of ten draws, twenty eligible patients per class are sampled, and nested cohorts of five patients with 32 patches, ten with sixteen, and twenty with eight are formed, so every classifier trains on 160 patches per class at a single patient count for all classes. No additional patients, labels, features, or foundation models enter, and the reused arms are the exact fits of the source studies on the same cohorts, so every contrast is paired draw by draw.

A pre-fit gate tested whether the within-patient complement locates the missing span at all, before any classifier was fitted. On split 0, for ten draws at five patients, it measured the share of the pool’s between-patient variance that lies in the span of the first k complement directions, for $k \in \{r_B, 2r_B, 4r_B, 8r_B\}$, and compared it with k random complement directions and with the best k -dimensional subspace of the complement, which consists of the leading eigenvectors of $\mathbf{P}_\perp \Sigma_{\text{pool}} \mathbf{P}_\perp$. The gate fraction $f = (W - \text{random}) / (\text{oracle} - \text{random})$ is zero for directions no better than random and one for the best possible directions. The rule fixed in advance was to skip all fits if $f < 0.1$ at every k .

Ten new arm families enter at each $G \in \{5, 10, 20\}$ (Table 1). Every fitted arm whitens by Equation 2 with its expanded basis and spectrum and then fits an ordinary multinomial logistic classifier on the transformed features. It tunes the scale factor $\kappa \in \{0.1, 1, 10, 100\}$, rescaled as described in Section 2, jointly with the regularization strength on validation patient-macro balanced accuracy, with the grid and tie rules of Queisler (2026e). The tuned families Wt and Rt fit nothing of their own. For each fit they read the validation score that each fixed- s arm reached at its selected κ , add the score of the reused cohort-whitening fit as the $s = 0$ candidate, and copy the winning record. Ties go to the larger s . The random directions are drawn from a generator seeded by split, draw, and patient count, so the control is reproducible and differs between cohorts.

The eigen-decompositions are computed once per cohort and patient count and shared by the

Arm	Added directions	Added trace	Role
RG	—	—	Ordinary fit on the real cohort (reused)
CG	—	—	Cohort with pool class centres (reused headroom reference)
$RWcG$	none	0	Cohort whitening, the host remedy and $s = 0$ (reused)
RWG	pool basis	pool	Pool-basis whitening (reused mechanism reference)
WaG – WdG	within-patient $\mathbf{V}$	$s \in \{0.1, 0.3, 1, 3\}$	Expanded cohort whitening, Equation 4
WtG	within-patient $\mathbf{V}$	tuned	s selected on validation from $\{0, 0.1, 0.3, 1, 3\}$
RaG – RdG	random, $\perp \mathbf{U}$	$s \in \{0.1, 0.3, 1, 3\}$	Direction-specificity control, same eigenvalues
RtG	random, $\perp \mathbf{U}$	tuned	Control with s selected like Wt

Table 1: The W and R families keep the cohort basis and its spectrum and add directions orthogonal to it; they differ from each other only in which directions carry the added eigenvalues. RG and CG bracket the pool-centre headroom; $RWcG$ and RWG bracket what whitening achieves from the cohort and from all training patients. RWG reads the training pool and is not a remedy.

arms that use them. Per dataset, the study fits eight families at four scale factors for each of three patient counts and 30 split-draw pairs, that is 2,880 whitened fits, each tuning nine regularization values; the tuned families add no fit. Selections were written to disk before any test prediction, and test accuracy never entered selection.

4 Evaluation

The endpoint is patient-macro balanced accuracy on the test partition, pooled over splits and draws with equal weight per split. Uncertainty combines Bayesian bootstrap weights on test patients (Rubin, 1981) with resampling of draws within each split over 2,000 replicates; all arms share the replicates, so every contrast is paired, and intervals are 95 percent percentile intervals.

The primary contrast is $Wt5$ – $RWc5$, read separately on each dataset. The within-patient complement counts as a practical success on a dataset if the estimated gain over cohort whitening is at least one percentage point and its paired interval lies entirely above zero. The specificity contrast is $Wt5$ – $Rt5$. The *headroom share* $(WtG - RWcG)/(RWG - RWcG)$ states how much of the gap between cohort-basis and pool-basis whitening the added directions close. As in the preceding reports, the recovery $(WtG - RG)/(CG - RG)$ of the pool-centre headroom is reported at five and ten patients, and the *gap share* $1 - (Wt20 - Wt5)/(R20 - R5)$ states how much of the five-to-twenty patient gap the arm removes.

Three diagnostics, computed after fitting, relate accuracy to the span. The *captured share* is the fraction of the pool’s between-patient variance in the span of the top k eigenvectors of the selected $\hat{\Sigma}_s$, for $k \in \{r_B, 2r_B, 4r_B, 8r_B\}$; with $s = 0$ it equals the cohort basis’s share for every k . The *added overlap* is the subspace overlap $\|\mathbf{A}\mathbf{Q}^\top\|_F^2/r_B$ between the first r_B added directions $\mathbf{A}$ and the top r_B pool directions $\mathbf{Q}$ (Queisler, 2026f). Finally, the per-draw correlation between the gain in captured share and the accuracy gain Wt – RWc asks whether draws that capture more of the pool also gain more. Three readings were fixed in advance. $Wt \gg RWc$ together with $Wt \gg Rt$ would mean that the patch residuals supply patient directions the means miss. $Wt \approx Rt \gg RWc$

would mean that contracting more dimensions helps regardless of direction. $Wt \approx RWc$ despite a passed gate would mean that locating the missing span is not sufficient.

5 Results

The results are presented in the order of the argument: the pre-fit gate, the accuracy of the expanded arms against cohort whitening and the random control, the profile over the added trace s , and the diagnostics that relate accuracy to the captured span.

5.1 Pre-fit gate

The within-patient complement locates the missing span well on both datasets, and the gate passed by a wide margin (Table 2). On TCGA-UT its leading $r_B = 120$ directions hold 14.2 percent of the pool’s between-patient variance at five patients, against 20.3 percent for the best 120-dimensional subspace of the complement and 1.5 percent for 120 random complement directions, so the gate fraction is 0.68. The fraction rises with k and reaches 0.93 at $k = 8r_B = 960$, where the within-patient directions hold 28.5 of the 30.2 percent that lies outside the cohort basis at all. On BRACS the fraction is lower, 0.56 at $k = r_B = 28$ and 0.73 at $k = 8r_B$, but it is still far above the stopping threshold of 0.1. The numbers are internally consistent: the oracle share never exceeds the complement’s share of the pool variance, and the within-patient share lies between the random and the oracle share at every k .

	TCGA-UT, $r_B = 120$				BRACS, $r_B = 28$			
	$k = r_B$	$2r_B$	$4r_B$	$8r_B$	$k = r_B$	$2r_B$	$4r_B$	$8r_B$
Within-patient complement	0.142	0.196	0.245	0.285	0.158	0.213	0.267	0.317
Random complement	0.015	0.030	0.059	0.118	0.005	0.009	0.019	0.037
Best complement subspace	0.203	0.249	0.281	0.298	0.279	0.348	0.398	0.420
Gate fraction f	0.68	0.76	0.84	0.93	0.56	0.60	0.66	0.73

Table 2: The within-patient complement finds most of the pool’s between-patient variance that the cohort basis misses. Entries are shares of the pool’s between-patient variance in the span of the first k complement directions, at five patients per class, averaged over ten draws of split 0; the gate fraction is $(W - \text{random})/(\text{oracle} - \text{random})$, averaged per draw. The cohort basis itself holds 0.698 (TCGA-UT) and 0.577 (BRACS) of the pool variance (Queisler, 2026f).

5.2 Accuracy against cohort whitening

The expanded arm fails the prespecified criterion on both datasets. On TCGA-UT the tuned arm gains 0.45 points over cohort whitening at five patients per class (0.27 to 0.63), so the interval lies above zero but the estimate falls short of the one-point threshold (Table 3). On BRACS the gain has the same point estimate, 0.45 points, but its interval runs from -0.35 to 1.21 . The gain shrinks with the patient count: on TCGA-UT it is 0.06 points at ten patients (-0.05 to 0.19) and 0.00 at twenty (-0.03 to 0.03), and on BRACS 0.29 (-0.19 to 0.90) and -0.03 (-0.46 to 0.39).

The gain on TCGA-UT is specific to the within-patient directions. The tuned random control gains 0.02 points over cohort whitening at five patients (-0.01 to 0.06), and the specificity contrast $Wt5-Rt5$ is 0.42 points (0.25 to 0.60). Every fixed- s random arm stays within 0.04 points of cohort whitening at every patient count on TCGA-UT and within 0.07 points on BRACS; only two of these 24 contrasts have an interval that excludes zero, both small and negative: $Rc10$ and $Rd10$ on TCGA-UT at -0.01 (-0.02 to -0.00) and -0.04 points (-0.09

Arm family	$G = 5$	$G = 10$	$G = 20$
<i>TCGA-UT</i>			
Real cohort (R)	62.48 [61.24, 63.65]	67.79 [66.47, 69.00]	71.89 [70.55, 73.06]
Cohort whitening, $s = 0$ (RWc)	64.74 [63.52, 65.90]	69.64 [68.34, 70.84]	73.16 [71.86, 74.34]
$s = 0.1$ (Wa)	65.01 [63.79, 66.16]	69.69 [68.40, 70.88]	73.17 [71.87, 74.35]
$s = 0.3$ (Wb)	65.25 [64.02, 66.40]	69.71 [68.41, 70.86]	73.17 [71.87, 74.34]
$s = 1$ (Wc)	65.16 [63.91, 66.32]	69.65 [68.37, 70.85]	73.15 [71.84, 74.31]
$s = 3$ (Wd)	64.77 [63.50, 65.93]	69.50 [68.22, 70.72]	73.09 [71.79, 74.26]
Tuned s (Wt)	65.19 [63.95, 66.34]	69.71 [68.42, 70.89]	73.16 [71.87, 74.33]
Random complement, tuned s (Rt)	64.77 [63.54, 65.93]	69.65 [68.33, 70.83]	73.15 [71.85, 74.33]
Pool-basis whitening (RW)	67.40 [66.14, 68.52]	71.66 [70.39, 72.84]	74.44 [73.10, 75.59]
Pool centres (C)	73.14 [71.62, 74.50]	74.47 [73.01, 75.78]	75.55 [74.11, 76.85]
Gain Wt – RWc	0.45 [0.27, 0.63]	0.06 [−0.05, 0.19]	0.00 [−0.03, 0.03]
Gain Wt – Rt	0.42 [0.25, 0.60]	0.06 [−0.05, 0.18]	0.01 [−0.02, 0.03]
Gain Rt – RWc	0.02 [−0.01, 0.06]	0.00 [−0.01, 0.01]	−0.01 [−0.02, 0.01]
Gain Wt – R	2.71 [2.35, 3.04]	1.92 [1.62, 2.20]	1.27 [1.04, 1.50]
<i>BRACS</i>			
Real cohort (R)	36.18 [33.46, 38.80]	38.49 [35.87, 41.22]	40.97 [38.50, 43.62]
Cohort whitening, $s = 0$ (RWc)	36.49 [33.87, 39.24]	38.61 [35.94, 41.34]	41.28 [38.70, 43.88]
$s = 0.1$ (Wa)	37.09 [34.47, 39.70]	38.69 [36.03, 41.43]	41.20 [38.64, 43.74]
$s = 0.3$ (Wb)	37.14 [34.59, 39.70]	39.02 [36.33, 41.65]	41.32 [38.77, 43.89]
$s = 1$ (Wc)	36.91 [34.28, 39.52]	38.99 [36.32, 41.65]	41.24 [38.73, 43.78]
$s = 3$ (Wd)	36.77 [34.14, 39.35]	38.80 [36.19, 41.42]	41.26 [38.77, 43.78]
Tuned s (Wt)	36.94 [34.39, 39.48]	38.90 [36.30, 41.53]	41.26 [38.76, 43.77]
Random complement, tuned s (Rt)	36.56 [33.91, 39.27]	38.65 [35.96, 41.40]	41.26 [38.72, 43.81]
Pool-basis whitening (RW)	39.50 [36.82, 41.98]	40.13 [37.39, 42.77]	41.25 [38.79, 43.76]
Pool centres (C)	44.63 [41.33, 48.05]	45.11 [41.85, 48.51]	45.50 [42.20, 48.76]
Gain Wt – RWc	0.45 [−0.35, 1.21]	0.29 [−0.19, 0.90]	−0.03 [−0.46, 0.39]
Gain Wt – Rt	0.38 [−0.42, 1.16]	0.25 [−0.24, 0.86]	−0.01 [−0.37, 0.44]
Gain Rt – RWc	0.06 [−0.03, 0.26]	0.04 [−0.04, 0.16]	−0.02 [−0.28, 0.14]
Gain Wt – R	0.77 [−0.16, 1.72]	0.40 [−0.24, 1.13]	0.28 [−0.49, 1.15]

Table 3: Adding the within-patient complement to cohort whitening gains about half a point at five patients and nothing at twenty; the same eigenvalues on random directions gain nothing. Accuracy entries are test patient-macro balanced accuracy in percent; gains are paired contrasts in percentage points; bold entries are the primary contrast. Intervals are 95% bootstrap intervals over test patients and draws. The R, C, RWc, and RW rows are the fits of Queisler (2026b), Queisler (2026e), and Queisler (2026a).

to -0.01). Contracting random directions of the complement therefore neither helps nor hurts by more than a few hundredths of a point. On BRACS the specificity contrast is 0.38 points (-0.42 to 1.16), in the same direction as on TCGA-UT but not resolved.

Measured against the gap it was meant to close, the gain is small. At five TCGA-UT patients the tuned arm closes a share of 0.169 (0.105 to 0.231) of the 2.66-point gap between cohort-basis and pool-basis whitening, and 0.032 (-0.024 to 0.089) of the 2.02-point gap at ten patients. Its recovery of the pool-centre headroom is 0.254 (0.221 to 0.289) at five patients and 0.287 (0.241 to 0.337) at ten, against 0.212 and 0.277 for cohort whitening. Because the gain is concentrated at five patients, it also narrows the patient-count gap: the tuned arm removes a share of 0.153 (0.119 to 0.185) of the 9.41-point gap between five and twenty patients, that is 1.44 points, against 0.105 (0.074 to 0.136) or 0.99 points for cohort whitening. On BRACS the headroom share is 0.149 (-0.140 to 0.391) of a 3.01-point gap at five patients, and the recovery of 0.091 (-0.020 to 0.196) is indistinguishable from zero.

5.3 The profile over the added trace

At five TCGA-UT patients the gain has an interior maximum in s (Figure 1). It is 0.26 points at $s = 0.1$ (0.14 to 0.40), 0.51 at $s = 0.3$ (0.33 to 0.68), 0.42 at $s = 1$ (0.21 to 0.63), and 0.02 at $s = 3$ (-0.17 to 0.23). The tuned arm, at 0.45 points, reaches nearly the best fixed value, so validation selection loses little. At ten and twenty patients no fixed s gains more than 0.06 points, and the largest added trace becomes harmful: $s = 3$ costs 0.14 points at ten patients (-0.27 to 0.00) and 0.07 at twenty (-0.12 to -0.02). The random arms are flat at zero across the whole grid.

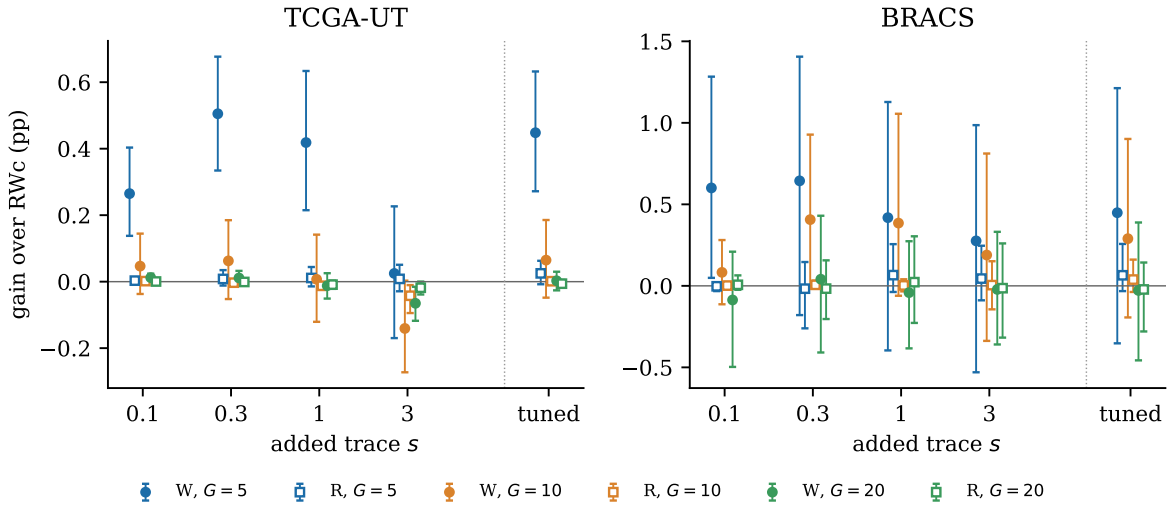


Figure 1: Within-patient directions help at five patients with an interior optimum near $s = 0.3$; the same eigenvalues on random directions do nothing at any s . Paired gain over cohort whitening in percentage points against the added trace s , as a fraction of the between-patient trace; $s = 0$ is cohort whitening and therefore zero by construction. Filled circles are the within-patient arms W, open squares the random-complement control R, and the markers right of the dotted line are the validation-tuned arms. Error bars show the paired 95% bootstrap intervals; markers are offset horizontally to separate patient counts and families.

The BRACS profile has the same shape at five patients but much wider intervals. Only the smallest added trace has an interval that lies above zero, 0.60 points at $s = 0.1$ (0.05 to 1.28); the peak estimate is 0.64 at $s = 0.3$ (-0.18 to 1.41). This is one of twelve fixed- s contrasts on BRACS and not the primary contrast, so it does not establish a gain on that dataset. At ten

patients the estimates are 0.08 to 0.41 points with intervals that include zero, and at twenty they are within 0.09 points of zero.

5.4 Diagnostics and validation selection

The selected expanded covariances capture much more of the pool’s between-patient variance than the cohort basis does, and far more than the random control (Table 4). At five TCGA-UT patients, the top r_B eigenvectors of the selected $\hat{\Sigma}_s$ capture 0.711 of the pool variance and the top $8r_B$ capture 0.977, against 0.698 for the cohort basis alone and 0.795 for the random control at $8r_B$. On BRACS the corresponding values are 0.583 and 0.866, against 0.577 and 0.607. The first r_B added directions overlap the pool’s top r_B directions by 0.27 on TCGA-UT and 0.19 to 0.22 on BRACS, five to twenty times the value $r_B/2,560$ of an unrelated subspace at five patients.

	TCGA-UT			BRACS		
	$G = 5$	$G = 10$	$G = 20$	$G = 5$	$G = 10$	$G = 20$
Rank $r_B = C(G - 1)$	120	270	570	28	63	133
Captured share, cohort basis	0.698	0.845	0.938	0.577	0.760	0.876
Captured share, Wt, $k = r_B$	0.711	0.842	0.911	0.583	0.737	0.844
Captured share, Wt, $k = 8r_B$	0.977	0.969	0.996	0.866	0.911	0.961
Captured share, Rt, $k = 8r_B$	0.795	0.926	0.985	0.607	0.798	0.912
Added overlap with pool basis	0.270	0.265	0.256	0.221	0.196	0.189
Overlap of a random subspace	0.047	0.105	0.223	0.011	0.025	0.052
Correlation with gain, $k = r_B$	0.35	0.15	0.33	0.37	0.08	−0.12
Correlation with gain, $k = 8r_B$	0.31	0.13	0.01	0.10	0.15	−0.05
Selected $s = 0$	0	6	2	2	7	7
Selected $s = 0.1$	3	5	8	2	5	1
Selected $s = 0.3$	12	10	9	7	3	8
Selected $s = 1$	13	8	8	6	5	8
Selected $s = 3$	2	1	3	13	10	6

Table 4: The expanded covariance captures most of the pool’s between-patient variance, yet per-draw capture predicts the accuracy gain only weakly. Captured shares are the pool variance in the span of the top k eigenvectors of the selected covariance; the cohort-basis row is taken from Queisler (2026f) on the same cohorts. The added overlap is that of the first r_B within-patient directions with the pool’s top r_B directions, reported for $s = 1$, and the random row is its reference value $r_B/2,560$. The correlation is taken across the 30 fits between the gain in captured share over the cohort basis and the test gain Wt–RWc. The first nine rows are means over 30 fits; the selection rows count the s chosen by Wt in those 30 fits.

Capture predicts accuracy only weakly. Across the 30 fits at five TCGA-UT patients, the correlation between the gain in captured share and the accuracy gain Wt–RWc is 0.31 to 0.35 at every k , and at ten patients it is 0.12 to 0.19. With 30 fits, a correlation of 0.35 lies at the edge of conventional significance (two-sided $p \approx 0.06$). At twenty patients the correlation falls to 0.01 at $k = 8r_B$, where the captured share is already close to one for every fit. The random control shows the same dissociation from the other side: it raises the captured share at $8r_B$ from 0.698 to 0.795 at five TCGA-UT patients without any accuracy gain.

Validation selection on TCGA-UT is informative at five patients and diffuse beyond. The tuned arm never keeps $s = 0$ at five patients and chooses $s = 0.3$ or $s = 1$ in 25 of 30 fits, which brackets the best fixed value. At ten and twenty patients the selections spread over the whole grid, in line with the flat profile there. The random control chooses $s = 3$ in 14 of 30 fits at five

TCGA-UT patients, which costs nothing because the random arms are flat. On BRACS both tuned families choose $s = 3$ most often at five patients, 13 and 14 of 30 fits, even though the test profile of the within-patient arm peaks at small s . This mismatch is what a noisy validation signal on 24 validation patients produces.

The whitening scale shifts with the added directions. On TCGA-UT at five patients the tuned arm selects grid factor 10 in 22 of 30 fits and factor 100 in 3, whereas cohort whitening selects factor 1 in almost all fits (Queisler, 2026e); the random control keeps factor 1 in 23 of 30. Because the factors are rescaled to act on the cohort block exactly as in cohort whitening (Section 2), a larger factor contracts both the cohort block and the added block more strongly. At ten patients the tuned arm selects factor 1 in 18 and factor 10 in 11 fits, and at twenty patients factor 1 in all 30. On BRACS the factors spread mostly from 0.1 to 10 in every family, as in the earlier BRACS studies (Queisler, 2026a).

6 Discussion

Of the three readings fixed in Section 4, the results select a weak form of the first on TCGA-UT and the third in substance. The within-patient complement locates the missing span, as the gate shows, and whitening along it helps in a direction-specific way at five patients, because the random control with identical eigenvalues gains nothing. The effect is small, however: 0.45 points, a sixth of the 2.66-point gap to pool-basis whitening, below the practical threshold, and absent at ten and twenty patients. On BRACS no contrast is resolved.

The central finding is a dissociation between span and accuracy. At five TCGA-UT patients the top 960 eigenvectors of the selected expanded covariance hold 98 percent of the pool’s between-patient variance, almost as much as the pool basis itself, yet the arm recovers only 17 percent of the accuracy that the pool basis delivers. Capture also predicts the per-draw gain only weakly. The captured share measures whether the between-patient variance lies in the span of the chosen directions, but the whitening in Equation 2 acts through the weight ν_j on each direction. In the expanded arm those weights are the patch-level variances $\sigma\tau\omega_j$, not the patient-level variances the pool basis uses. The within-patient complement ranks directions by how much patches vary within a patient. Along any single direction of this ranking, the between-patient variance is a fraction of the patch-level variance, and the class signal carried by the same direction is contracted along with the patient variation. A single scalar s can trade these effects against each other only on average. This reading fits the interior optimum near $s = 0.3$, the harm at $s = 3$ for larger patient counts, and the stronger scale factor selected at five patients. It is an interpretation, not a tested mechanism: the arm that would test it, the expanded basis with the pool’s between-patient variance attached to each added direction, was not run.

The result sharpens the conclusion of Queisler (2026f) in one respect and qualifies it in another. That study showed that the eigenvalues within the cohort basis hardly matter and that the missing directions account for almost all of the headroom. The present study shows that the missing directions can be found from the cohort’s own patches, but that finding them is not enough. Outside the cohort basis, the weight attached to each direction does matter, and the cohort has no patient-level estimate of that weight, because its $C(G - 1)$ patient deviations do not reach those directions. The within-patient covariance is a proxy for direction, not for patient-level strength.

For the question of this series, what recovers the patient-shortage deficit from the cohort alone, the answer does not change. At five TCGA-UT patients, the best cohort-only whitening gains 2.71 points over the real cohort, against 2.26 for cohort whitening and 4.92 for pool-basis whitening, and it removes 15 percent of the five-to-twenty patient gap. The remaining headroom requires patient-level information along directions that the labelled cohort cannot supply.

Plausible sources are unlabelled slides from further patients, related cohorts, or a population covariance carried by the feature extractor.

7 Limitations

The study does not separate direction from weight within the complement. The expanded arms attach patch-level variances to the added directions, and the random control changes the directions while keeping those variances. An arm that keeps the within-patient directions and attaches the pool’s between-patient variance along each of them, in the manner of the oracle spectrum of Queisler (2026f), would bound what reweighting the same directions can achieve. It was not run, so the interpretation in Section 6 is plausible but untested.

The added trace is a single scalar, and the grid $\{0.1, 0.3, 1, 3\}$ was fixed before fitting. A truncation of the complement to its leading directions, or a separate weight per direction, could behave differently; truncation was used only as a diagnostic. The scale factor κ is tuned jointly with s , and at five TCGA-UT patients the tuned arm selects a larger factor than cohort whitening. The gain therefore reflects the joint contraction of the cohort block and the added block, not the added block alone. The random control, which uses the same anchoring and the same grid and gains nothing, excludes the reading that a stronger contraction of the cohort block alone explains the gain; cohort whitening also had factor 10 available and rarely selected it.

The gate and the captured-share diagnostic measure variance in a span, which is not the quantity the classifier uses, as the weak correlations show. The gate was computed on split 0 only. On BRACS the complement has about a thousand directions rather than the full complement, because $CG(m - 1)$ is smaller than the feature dimension, and the random control spans a different subspace there than on TCGA-UT. All arms reuse the three evaluation splits of the source studies, so they share test patients with every earlier comparison in this series, and repeated use of these partitions inflates the chance that some arm appears to work by selection alone; the one BRACS contrast with an interval above zero, at $s = 0.1$, is of this kind. The results are conditional on frozen Virchow2 features, multinomial logistic regression, 160 patches per class, and the two cohorts studied.

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